Brain and Spinal Cord Embryology and Developmental Abnormalities

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Outlines

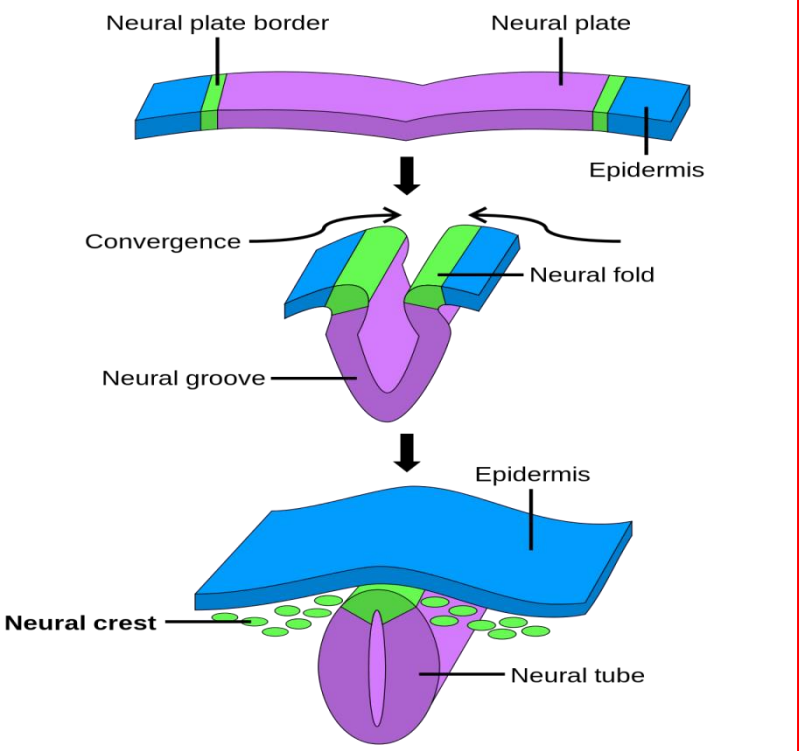
- Stages of CNS development.
- Neurulation and neural tube formation.
- Histogenesis of the neural tube.
- Nerve cells differentiation.
- Myelination.
- Neural crest cells.
- Neural tube defects (anomalies)

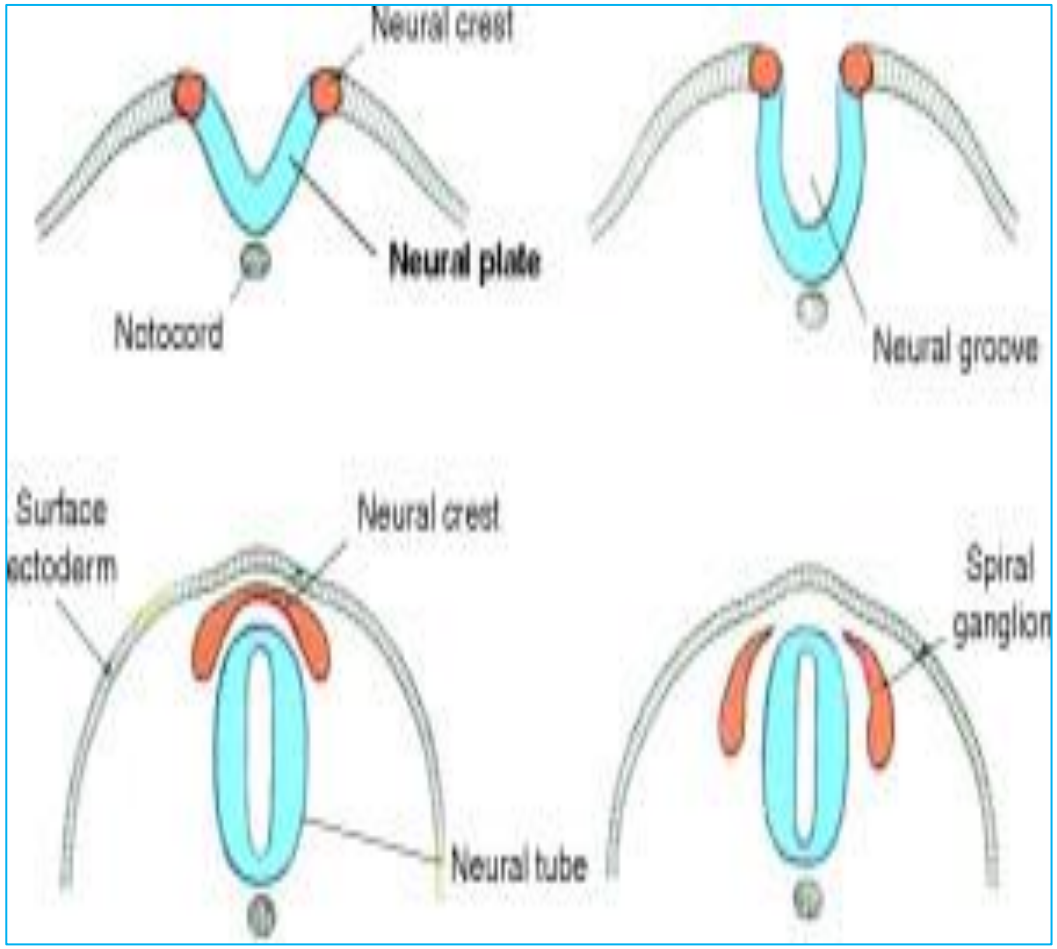
Stages of CNS Development

- Major stages of CNS development:
  1. Induction and neurulation.
  2. Separation and migration of the neural crest.
  3. Cellular proliferation and neuro blasts and oligo blasts migration .
  4. Axonal projection , dendritic proliferation and synaptogenesis.
  5. Fissure and sulci formation and myelination

Neural Tube

- The central nervous system (C.N.S) appears at the beginning of **3rd week** as slipper- shaped plate of thickened ectoderm, **the neural plate,** in the mid dorsal region in front of the ***primitive node***.
- Its lateral edges elevate to form the **neural folds,** which approach each other and fuse forming the **neural tube.**
- The fusion begins in **cervical** region and proceeds in cephalic and caudal direction.

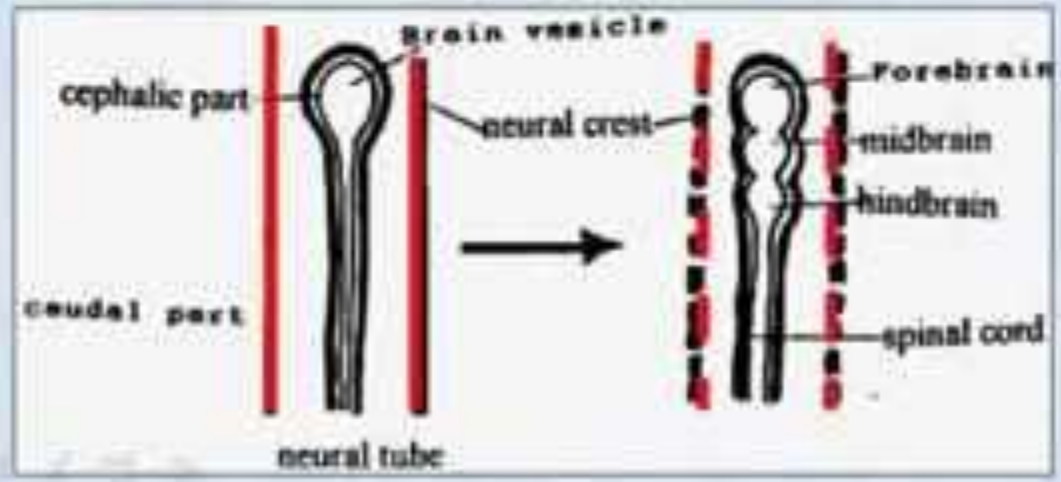

Neural plate border
Neural plate
Epidermis
Convergence
Neural fold
Neural groove
Epidermis
Neural crest
Neural tube

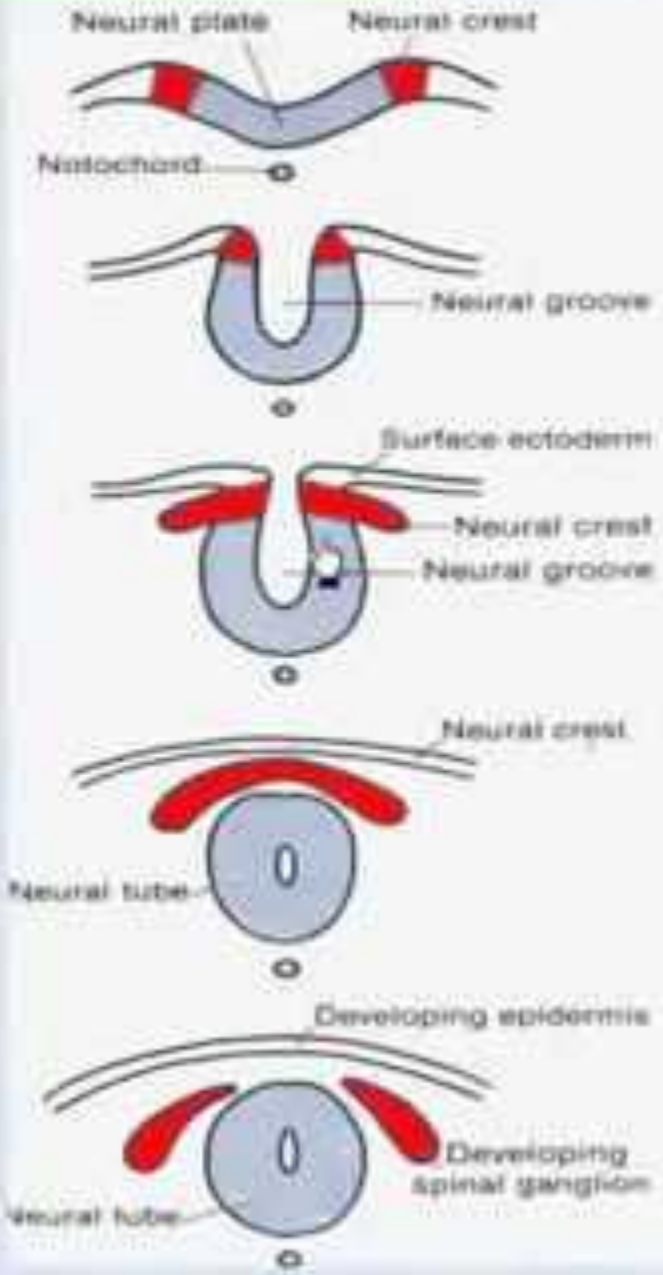

Neural crest
Neural plate
Notocord
Neural groove
Surface ectoderm
Neural crest
Neural tube
Spiral ganglion

DEVELOPMENT OF NEURAL TUBE

- The neural folds approach each other until the groove becomes converted into a "*neural tube*" which separates from the surface ectoderm.

- The neural tube is at first a simple tube. Later, its cranial end shows 2 constrictions and becomes divided into 3 vesicles: forebrain, midbrain and hindbrain vesicles.

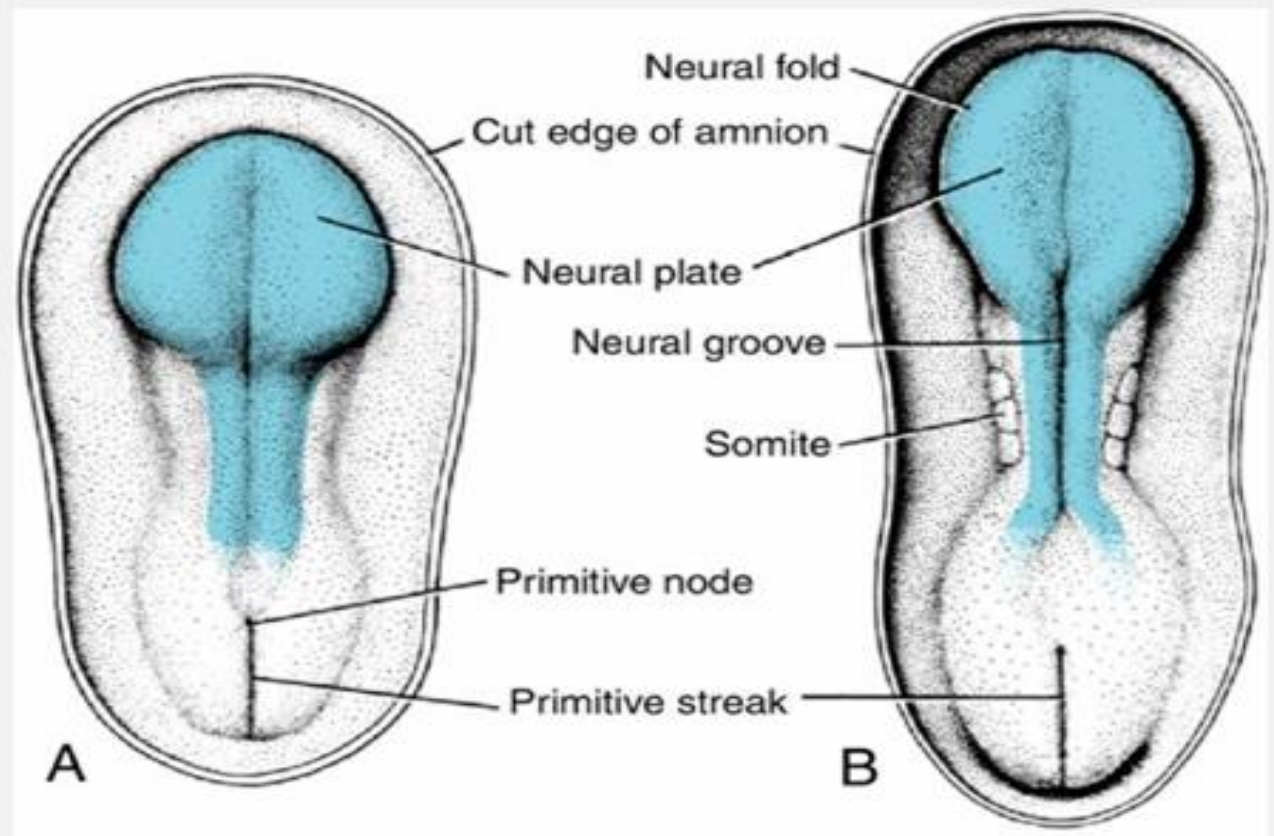




Neurulation

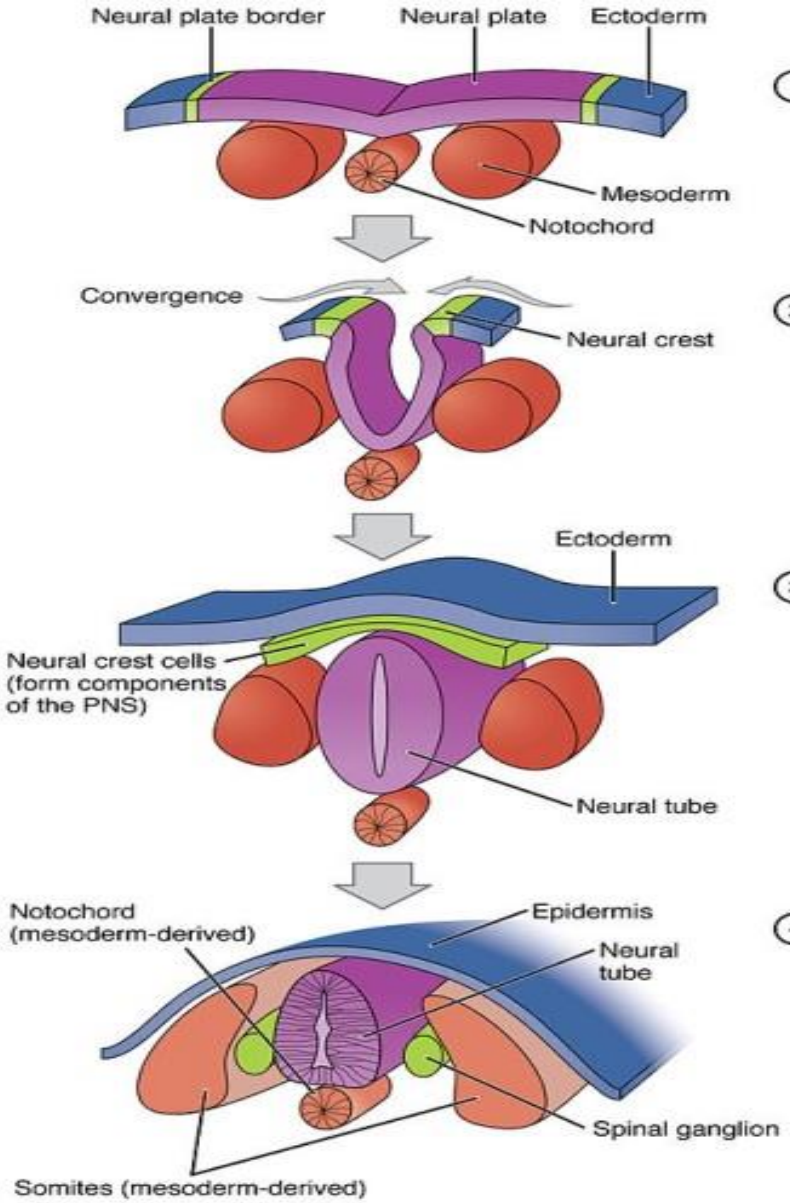
- The formation of the neural plate occur under inductive influence of the ***notochord.***
- At first, it is solid cord-like structure, then it become hollowed and gain a lumen.
- The process of neural tube formation can be described in two processes:
  1- **Primary neurulation** , the neural plate folds inward until the edges come in contact and fuse.
  2- **Secondary neurulation** , the tube forms by ***hollowing out*** of the interior of a solid precursor.

Neurulation



Dorsal view of the embryo showing the neural groove, neural folds and the somites

Stages of Neurulation



1. Neuroectodermal tissues differentiate from the ectoderm and thicken into the neural plate. The neural plate border separates the ectoderm from the neural plate.

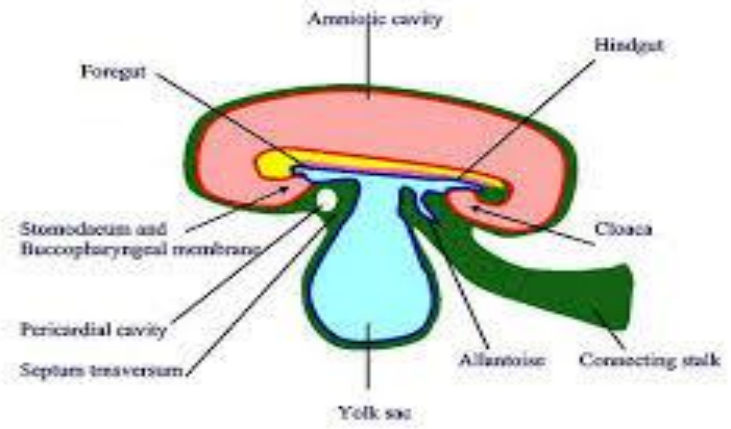
2. The neural plate bends dorsally, with the two ends eventually joining at the neural plate borders, which are now referred to as the neural crest.

3. The closure of the neural tube disconnects the neural crest from the epidermis. Neural crest cells differentiate to form most of the peripheral nervous system.

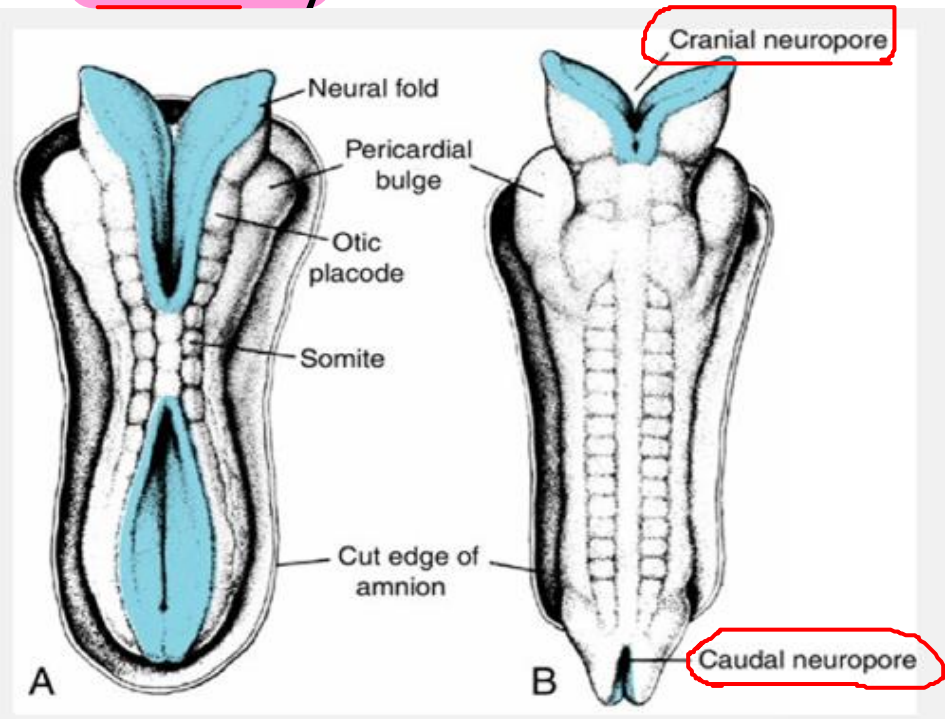
4. The notochord degenerates and only persists as the nucleus pulposus of the intervertebral discs. Other mesoderm cells differentiate into the somites, the precursors of the axial skeleton and skeletal muscle.

Neural Tube(cont.)

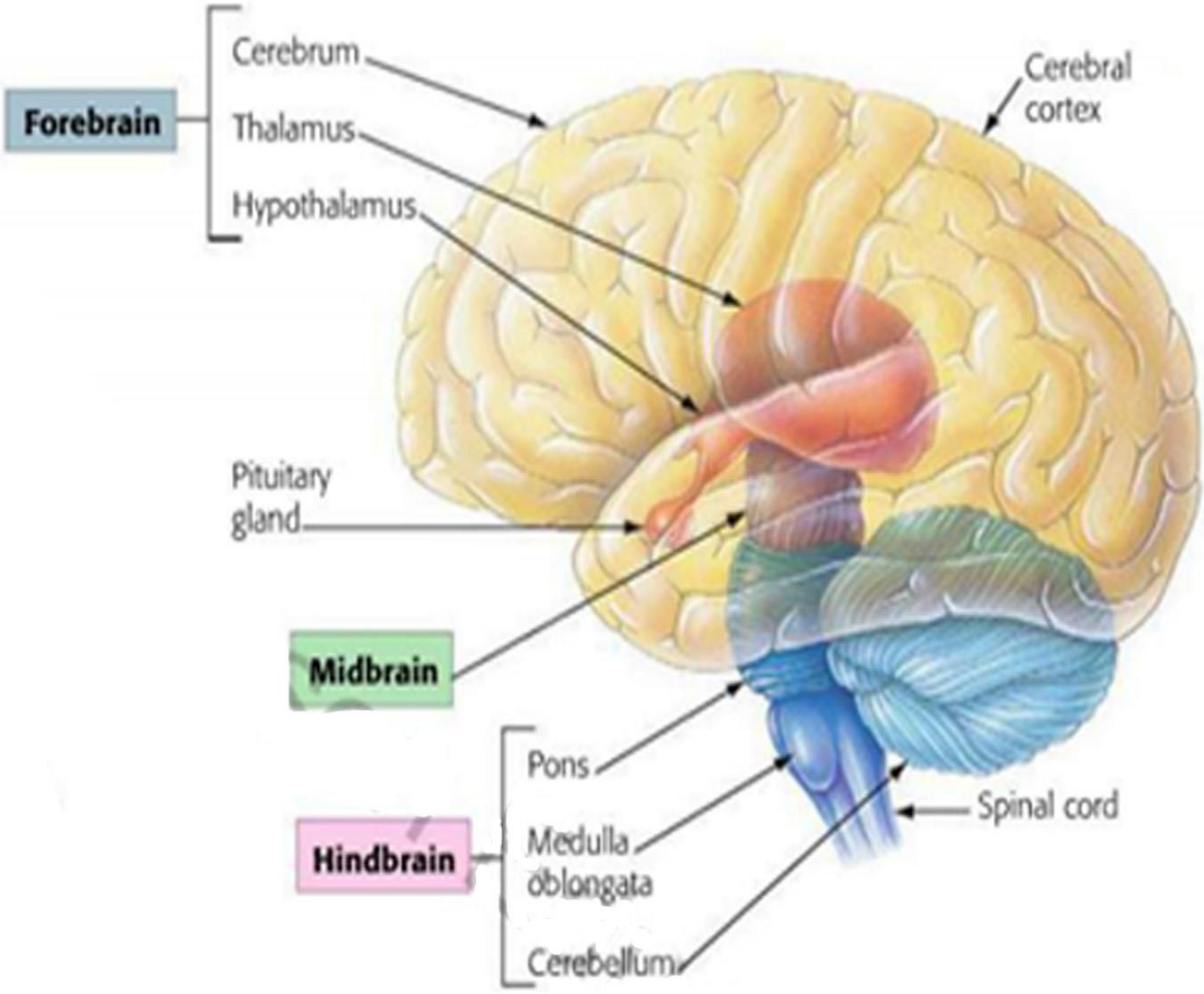
- The neural tube has a cephalic and a caudal neuropores that communicate with the overlying amniotic cavity.
- The cranial neuopore closes at the 25th day and the caudal neuropore closes at the 27th day .



Dorsal view of embryo showing the nervous system is connected to the amniotic cavity by the cranial and the caudal neuropores



The main parts of the human brain.



Neural Tube(cont.)

- The cephalic end (العلوي) of the neural tube shows three dilations, the primary brain vesicles:

1) Prosencephalon or forebrain, it consists of two parts:

1- Telencephalon (Cerebral hemispheres).

2- Diencephalon, characterized by outgrowth of optic vesicles.

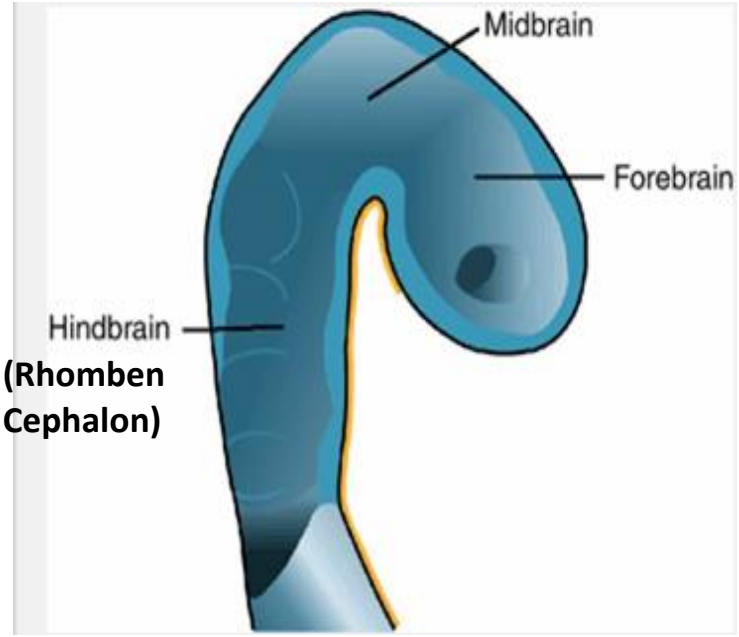
1) Mesencephalon or midbrain.

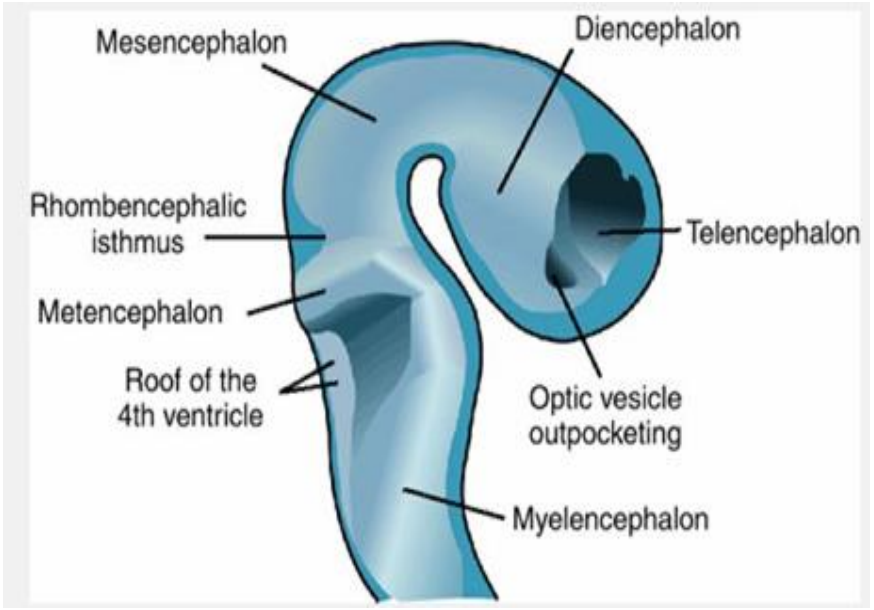
2) Rhombencephalon or hindbrain, also two parts:

1- Metencephalon :forms the pons and cerebellum.

2- Myelencephalon :forms the medulla oblongata.

Brain Vesicles

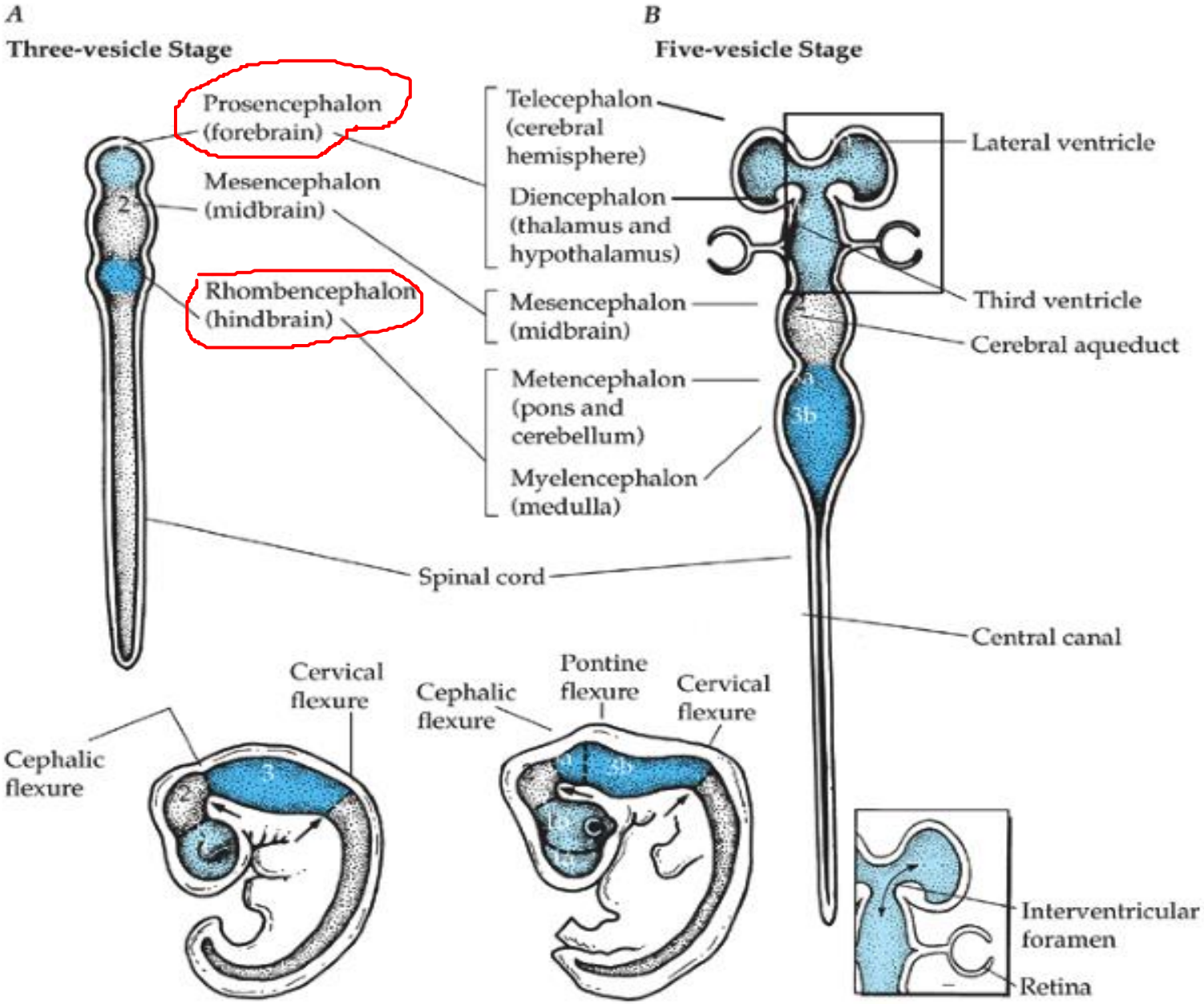




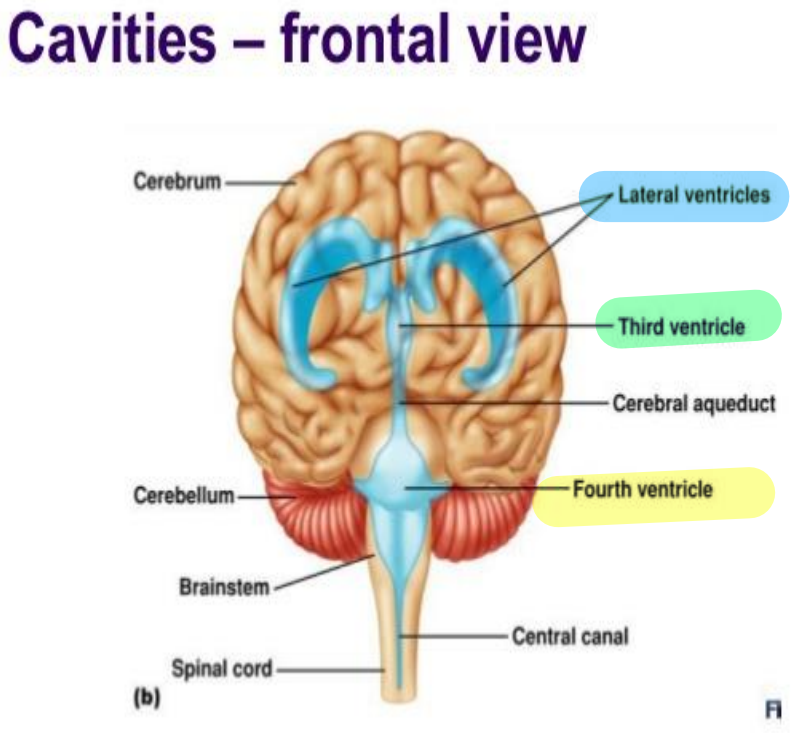
Primary brain vesicles

derivatives of the primary vesicles

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A
Three-vesicle Stage
Prosencephalon (forebrain)
Mesencephalon (midbrain)
Rhombencephalon (hindbrain)
Spinal cord
B
Five-vesicle Stage
Telecephalon (cerebral hemisphere)
Diencephalon (thalamus and hypothalamus)
Mesencephalon (midbrain)
Metencephalon (pons and cerebellum)
Myelencephalon (medulla)
Lateral ventricle
Third ventricle
Cerebral aqueduct
Central canal
Cervical flexure
Cephalic flexure
Pontine flexure
Cervical flexure
Cephalic flexure
Interventricular foramen
Retina

CNS cavities



The lumen of the spinal cord is the central canal continues with the brain vesicle.

- Rhombencephalon cavity is the 4th ventricle,
- deinencephaon cavity is the 3rd ventricle and
- these of cerebral hemispheres are the lateral ventricles.

Histogenesis of the Neural Tube

- At first, the wall of the neural tube is a simple layer of ***columnar epithelium***.
- The cells proliferates and form ***a pseudostratified neuroepithelium,*** which differentiated into **three** layers:

  1-Inner Ependymal layer → lining of central canal and ventricles.

  2-Middle Mantel layer → Grey matter.

  3-Outer Marginal layer : devoid of nerve cells and contains the nerve fibers of the cells in the mantel layer → White matter

Histogenesis of the Neural Tube

- Addition of neuroblasts to the mantle layer on each side of neural tube leads to formation of ventral and dorsal thickening.
- The ventral thickening forms the **basal plate** which forms the ventral motor horn cells.
- The dorsal thickening forms the **alar plate** which forms the sensory area of the tube .
- **Sulcus limitans** marks the boundary between the alar and the basal plats.

Histogenesis of the Neural Tube

- The ventral and dorsal midline portions of the tube become thin and form the **floor** and **roof plates**, contains no nerve cells and serve primarily as pathways for nerve fibers from one side to the other.

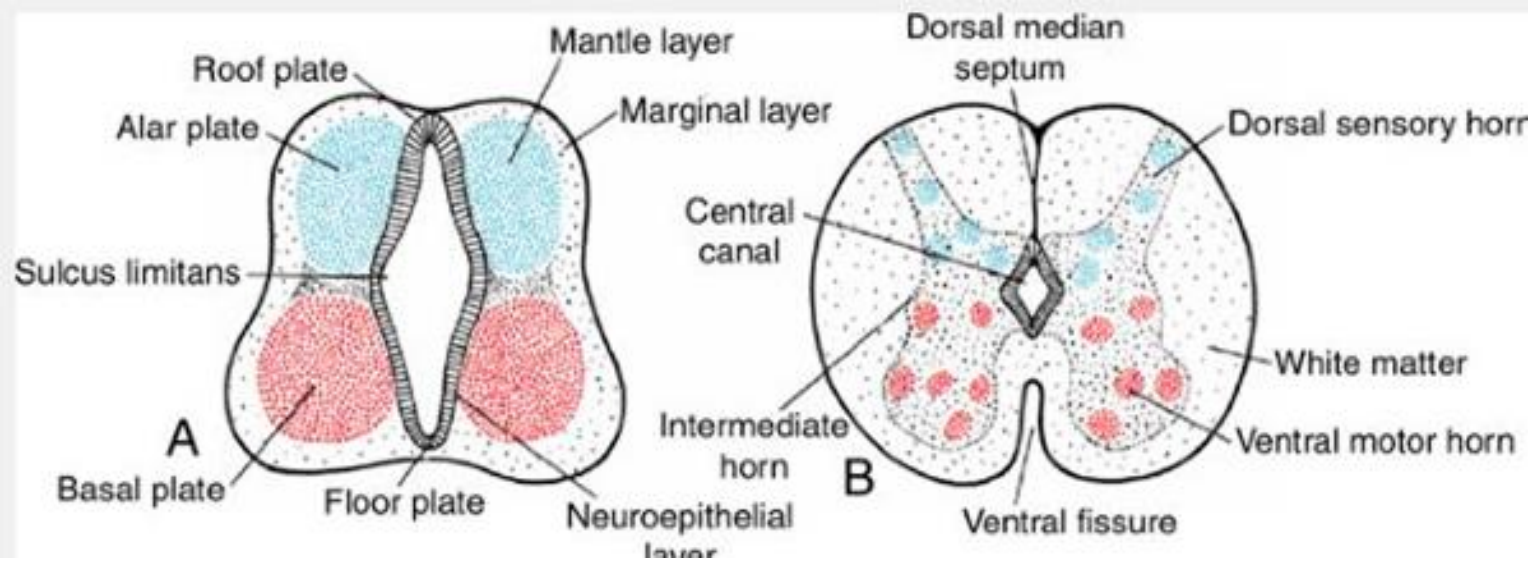


Figure showing formation of ventral motor dorsal sensory and intermediate column of the spinal cord.

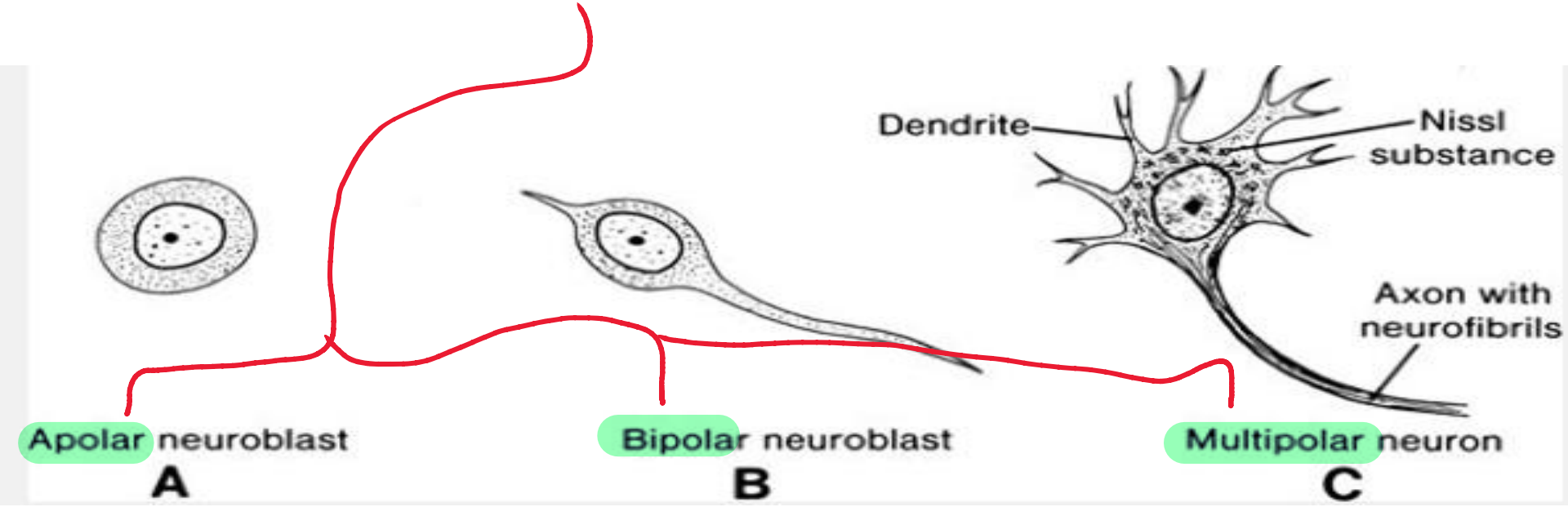
Histogenesis of the Neural Tube

middle layer

- The cells in the mantle layer are differentiated into two types:

  1- Neuro blasts.

Various stages of development of neuroblasts



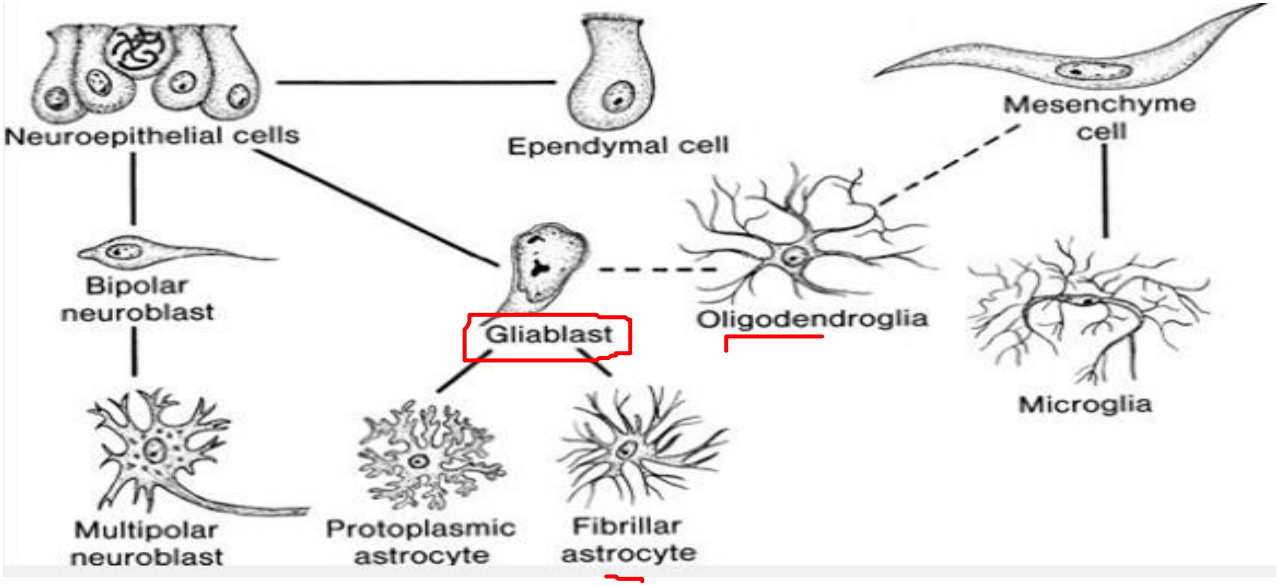
2- Glio blasts: differentiates into

a. Protoplasmic astrocytes.

b. Fibrillar astrocytes.

c. Oligodendroglial cells. Mainly in white matter and responsible for myelination of nerve fiber inside the C N S.

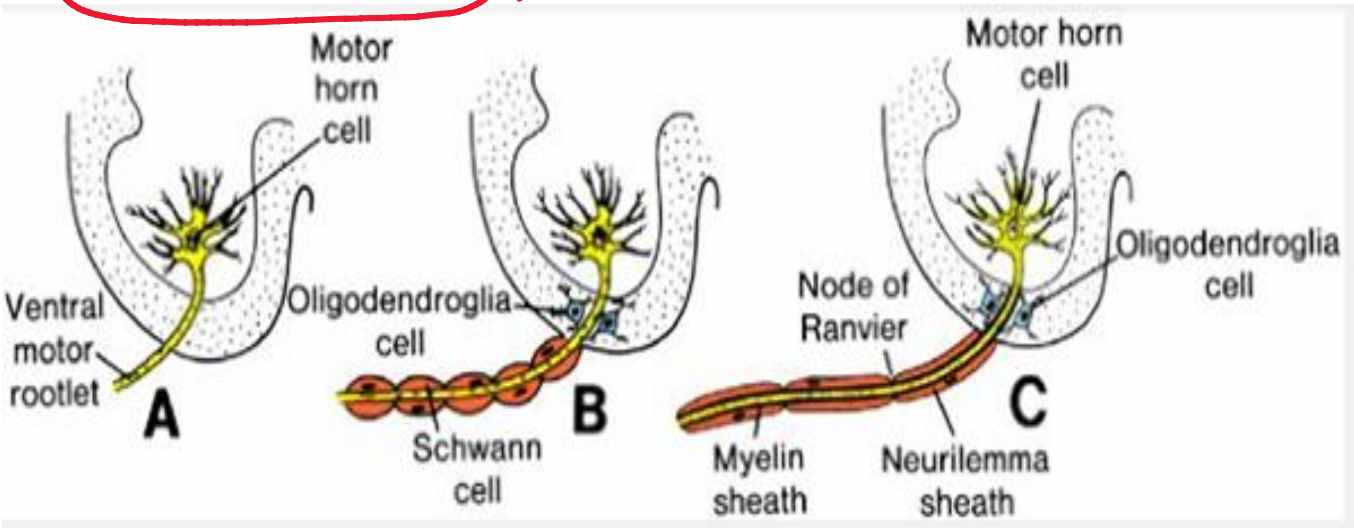
N.B Microglia are mesodermal in origin and (phagocytic)



Origin of nerve cells and various types of glial cells

Myelination of Nerve Fibers

- Starts at the 4th IU month and is not completed till the 2nd year after birth.
- Myelination of ***motor*** fibers occur earlier than sensory fibers.
- Myelination in CNS is by oligodendroglia while outside the CNS is by Schwann cells(from neural crest cells).



Neural Crest Cells

- Derived from the margin of the neural fold (ectodermal in origin) and migrate to give rise to:
  1. **All ganglia (**sensory and autonomic).
  2. **Liptomeninges** (Pia and arachnoid maters).
  3. Schwann cells and glial cells
  4. Melanocytes (melanin-producing cells) located in: the skin's epidermis, the middle vascular layer of the eye (the uvea).
  5. Chromaffin cells: adrenal medulla , carotid body and paraganglia.

6-Skeletal element: odontoblasts and cartilaginous elements of the ***branchial arches***.

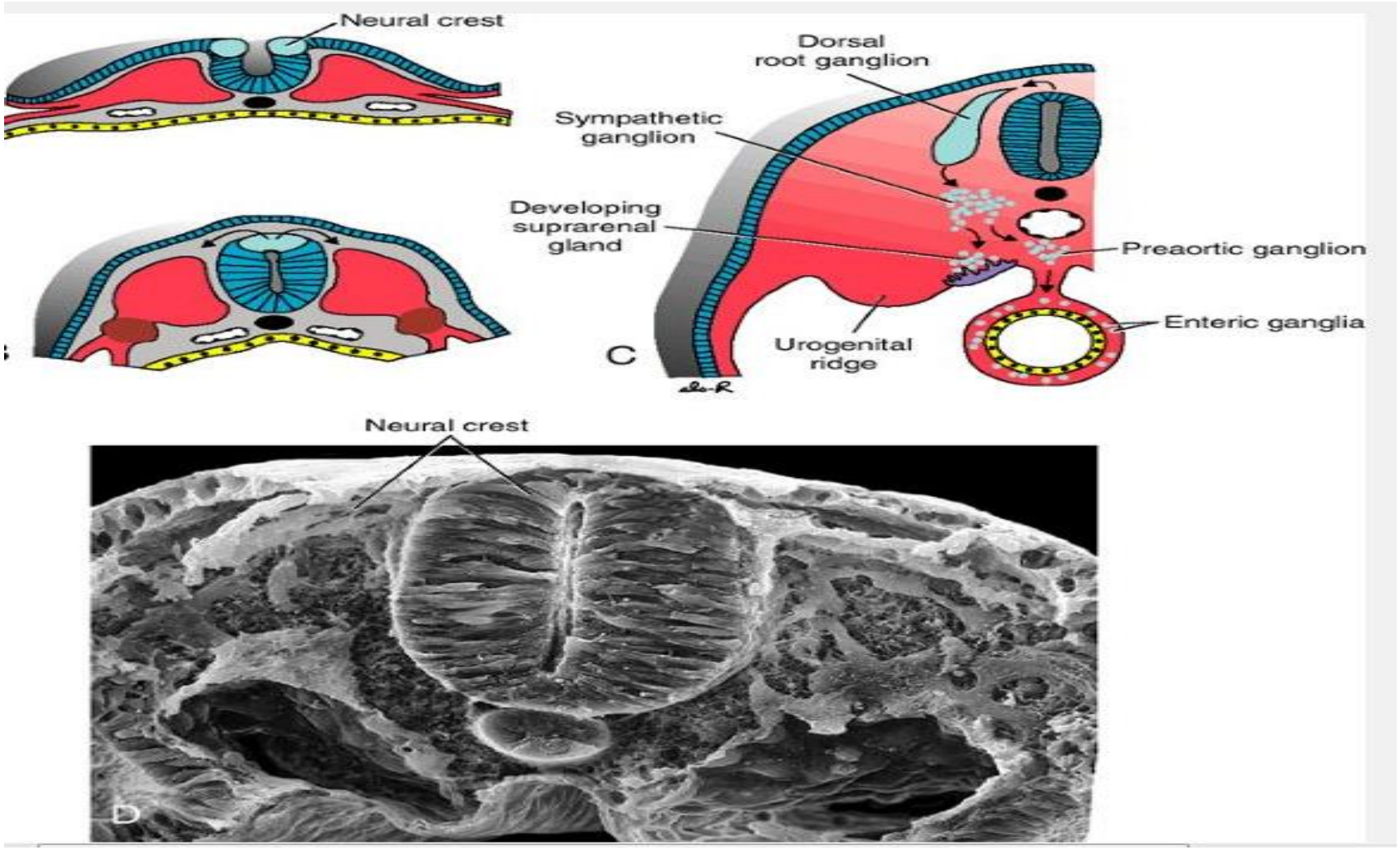
7-Chonotruncal(Aortico pulmonary) septum of the heart : separates the aorta and pulmonary arteries and fuses with the IV **septum** within the **heart** during **heart** development.

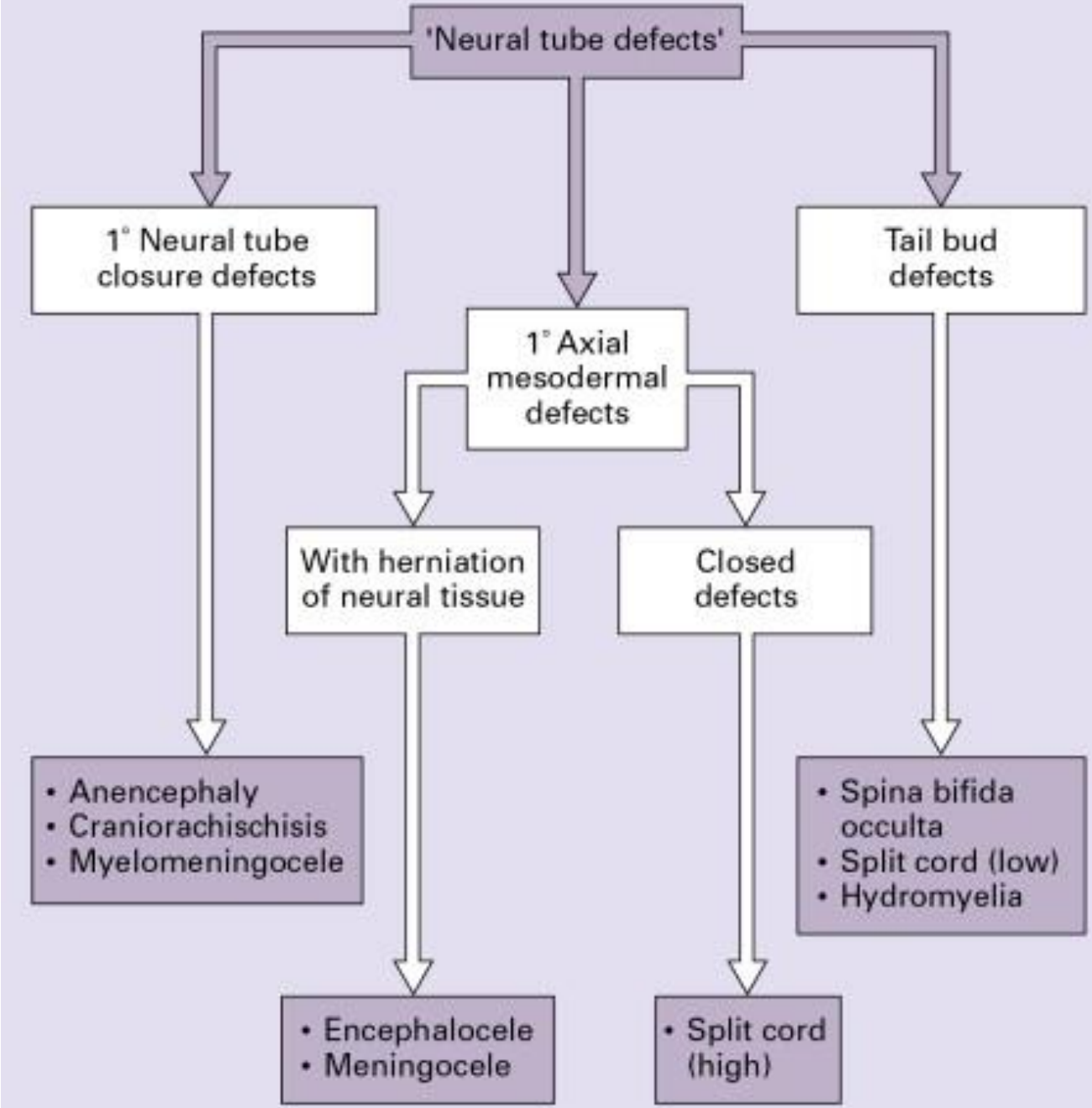
8- Smooth muscles to blood vessels of the face and forebrain.

9-Conective tissue and dermis of the face and neck.

10- C- cells of the thyroid gland.

Notice the migration Of the neural crest Cells to form Ganglia and Adrenal medulla




'Neural tube defects'
1° Neural tube closure defects
1° Axial mesodermal defects
Tail bud defects
With herniation of neural tissue
Closed defects
• Anencephaly
• Craniorachischisis
• Myelomeningocele
• Encephalocele
• Meningocele
• Split cord (high)
• Spina bifida occulta
• Split cord (low)
• Hydromyelia

Neural tube defects (NTDs): Group of birth defects in which an opening in the spinal cord or brain remains from early in human development. When the neural tube does not close completely, NTDs develop.

Specific types include:

1. Spina bifida which affects the spine.
2. Anencephaly which results in little to no brain.
3. Encephalocele which affects the skull.

NTDs are one of the most common birth defects, affecting over 300,000 births each year worldwide.



A- Anencephaly

= Absence of a major portion of the brain, skull, and scalp that results from a neural tube defect that occurs when the **cranial end** of the neural tube fails to close, between the 23rd - 26th day of IUL.



B-Anencephaly with complete Rachischisis (split spine) :

When the posterior neuropore of the neural tube fails to close by the 27th intrauterine day. So , the vertebrae overlying the open portion of the spinal cord do not fully form and remain unfused and open, leaving the spinal cord exposed.

C-Lumbosacral

Meylomeningocele

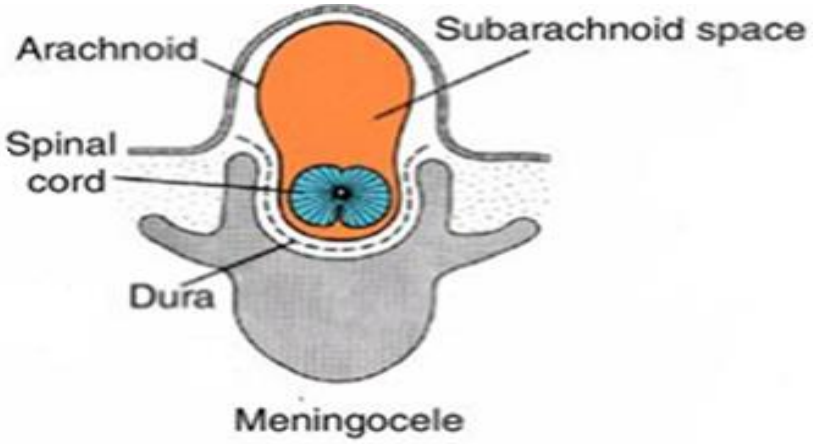


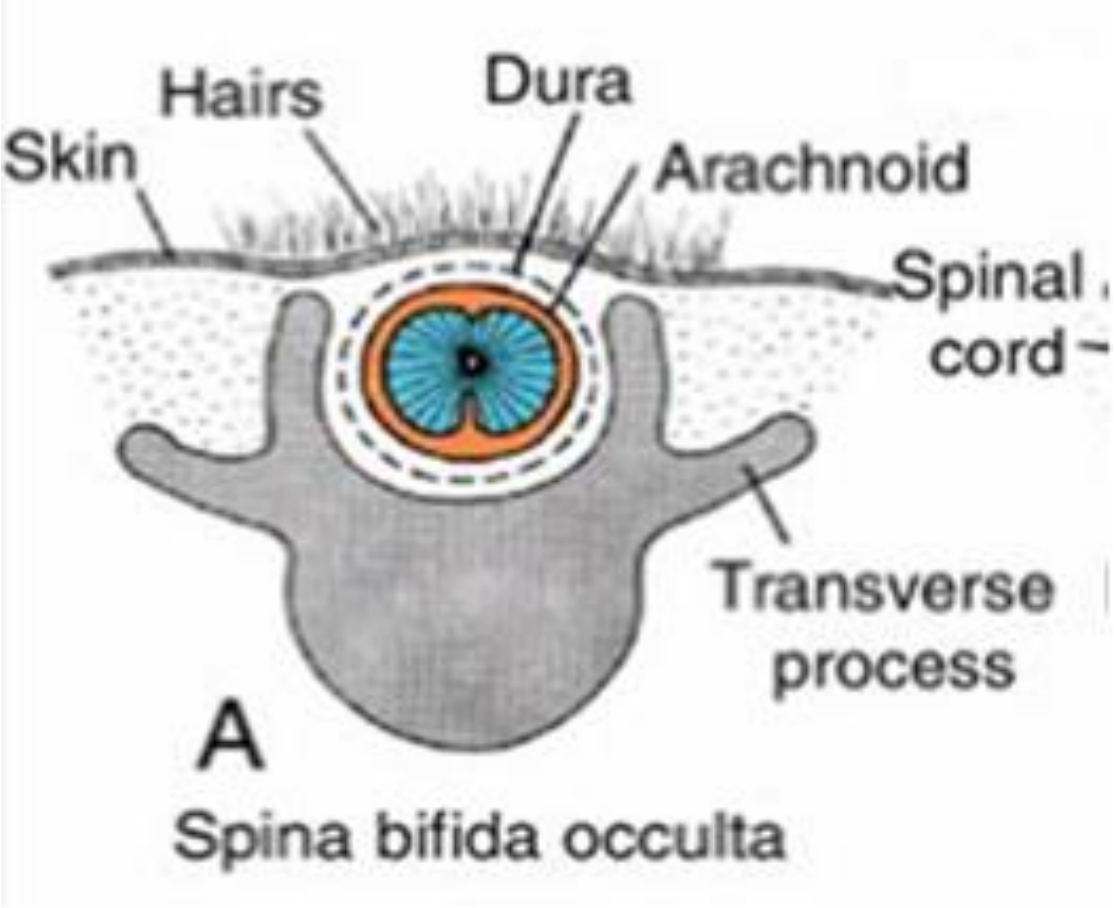


Occipital encephalocele



Anterior encephalocele

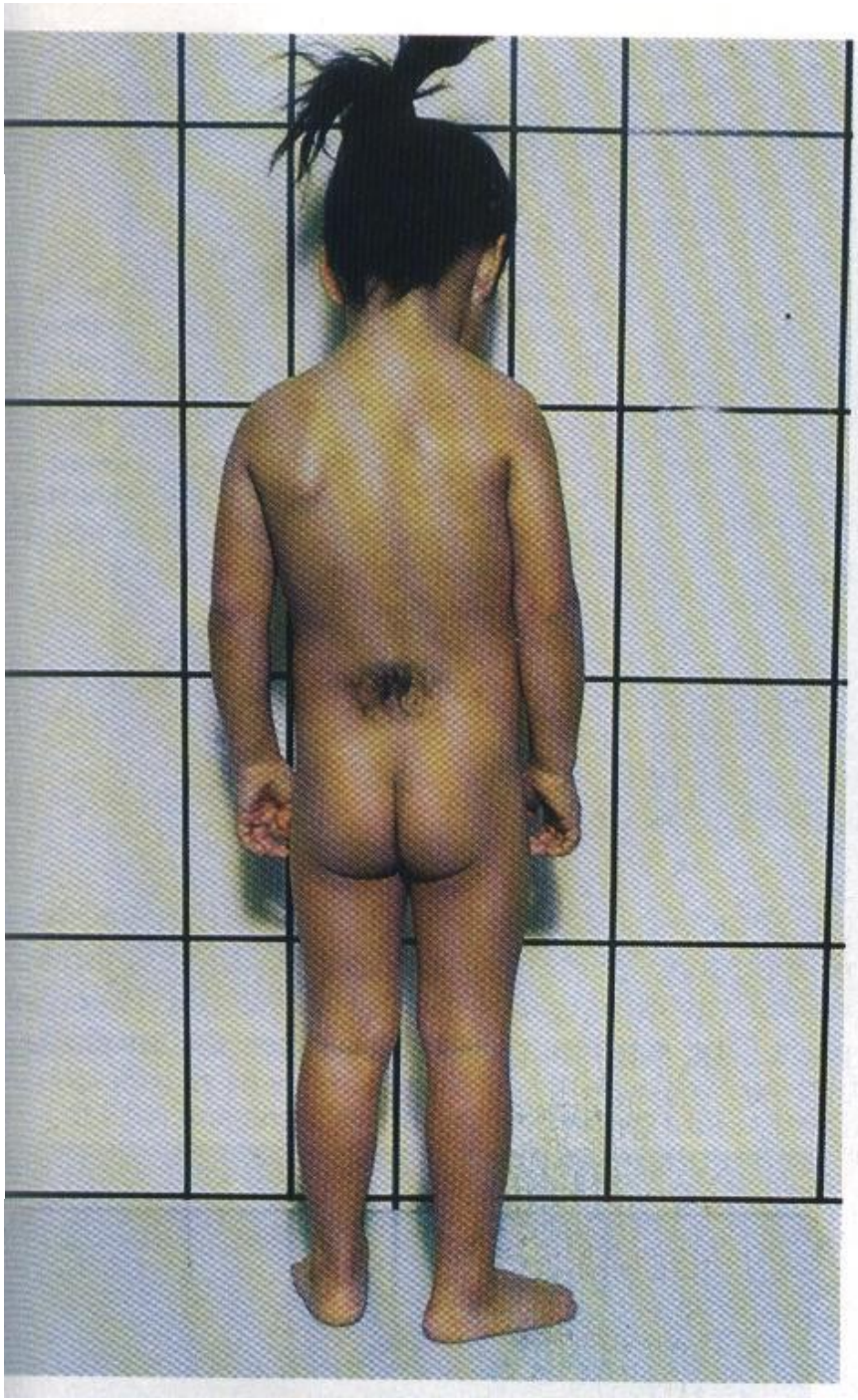


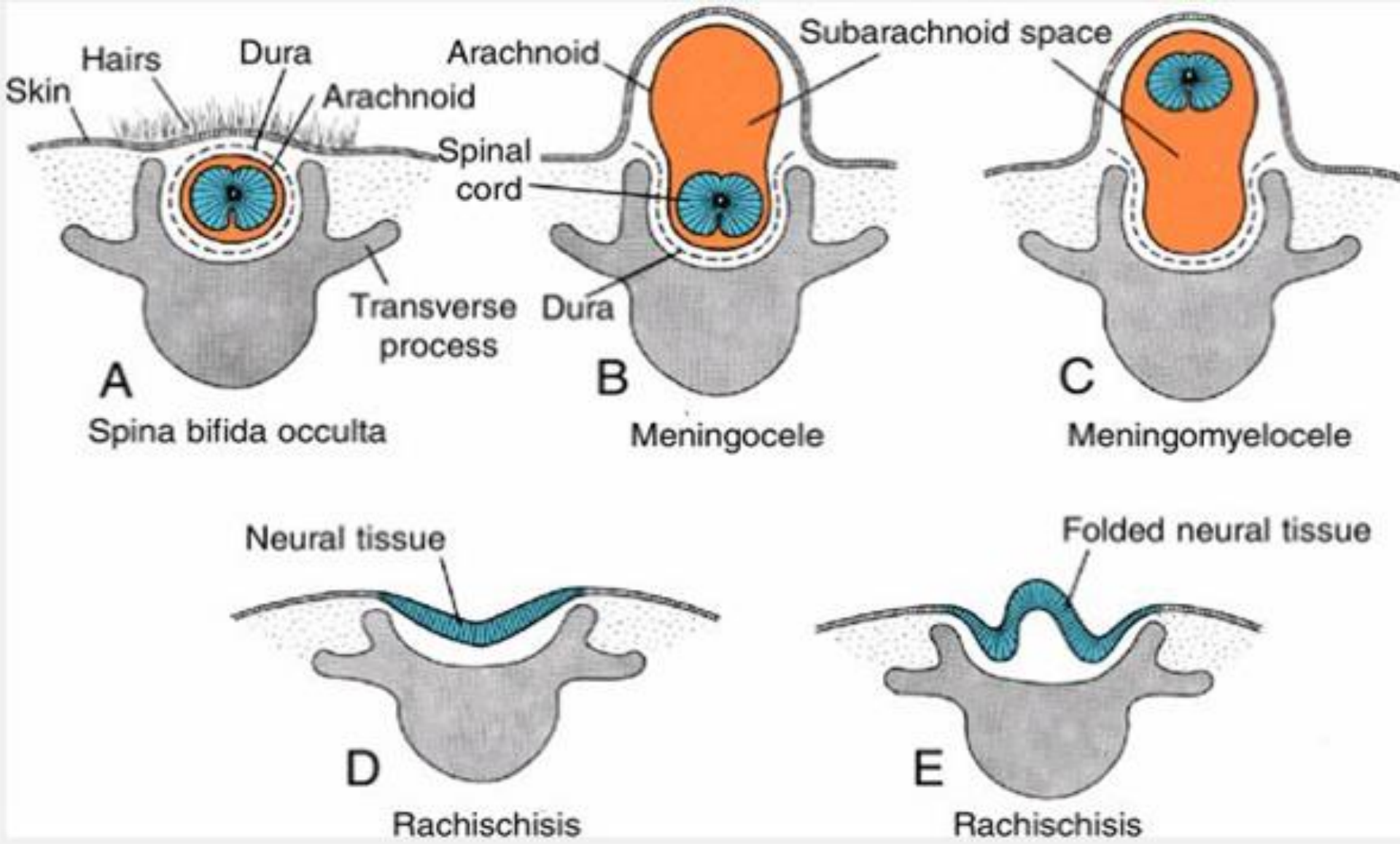


Defect of **tail bud development**: The vertebrae don't fully enclose the spinal cord.

The least severe group of neural tube defects :mostly in the low lumbar and sacral regions, broadly corresponding to the region of secondary neurulation


Spina Bifida (Open Defect)
Vertebra
Dura Mater
Spinal Cord
Spinal Fluid





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Summary

- The nervous system is entirely ectodermal in origin.
- It begins as neural plate, then neural groove which forms neural tube,
- The neural crest cells, differentiated into cells of the all ganglia, the cells of the adrenal medulla and the melanocytes. It gives also mesenchymal cells in the head and neck.
- The neural tube has cephalic and caudal neuropores which closes at the 25th and 27th days respectively.

Summary

- The cephalic end of the neural tube gives the primary brain vesicles: prosencephalon (forebrain), mesencephalon (midbrain) and rhombencephalon (hindbrain).
- The Prosencephalon consists of
  - Telencephalon, (cerebral hemispheres)
  - Diencephalon, (thalamus and hypothalamus)
- The Rhombencephalon consists of
  - Metencephalon (pons and cerebellum)
  - Myelencephalon (medulla oblongata)

Summary

- The lumen of the spinal cord is the central canal
- The cavity of the rhombencephalon is the fourth ventricle
- The cavity of the diencephalon is the third ventricle
- The cavity of the Telencephalon are the lateral ventricles.